

Null-Spinor Area as the Rest Gap of an Exactly Unitary Dirac Walk

A Machine-Verified Finite Construction

Mark Schwab

July 10, 2026

Abstract

The rest gap is normally an independent input to a Dirac quantum walk. Here it is fixed by supplied null-spinor geometry, with no second mass parameter, and we prove exactly what that derivation adds. For massless rank-one momenta $P = \sum_i \psi_i \psi_i^\dagger$, we prove

$$\mu^2 := \det P = \sum_{i < j} |\psi_i \wedge \psi_j|^2,$$

with $\mu = 0$ exactly when the occupied spinors span at most one complex direction. For two spinors, the complex area $z = \psi_1 \wedge \psi_2$ determines the odd Hermitian operator

$$B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \det(P)\mathbf{1}, \quad (k\sigma_z + B_z)^2 = (k^2 + \det P)\mathbf{1},$$

and every odd Hermitian rest operator is B_w for a unique $w \in \mathbb{C}$: the rest gap is the null-spinor area, with no independent mass parameter. Exponentiating B_z gives an exactly unitary walk with quasienergy $\cos(\omega a) = \cos(ka) \cos(\mu a)$, corner ratio $-i \tan(a\mu)$, and a directed history sum in which opposite turns carry z and $\bar{z}$. We prove that the walk's fixed-momentum symbols converge to the corresponding Dirac flow uniformly on bounded momentum boxes at rate $O(1/n)$, for the $1 + 1$ walk and for an ordered $3 + 1$ successive-axis walk with the complex phase retained; an arbitrary position-dependent Pluecker field $z(x)$ generates an exactly reversible finite-torus walk with a strict causal cone, site-dependent chiral rotation induces explicit endpoint links while a constant phase induces none, and compactly supported momentum-space data converge in L^2 at every fixed time. The link data derived from any nowhere-zero Pluecker field has an exact 2π -integer total turning, invariant under global chiral phase, with winding one realized by an explicit field generated by primitive spinors; the patched data is derived rather than supplied, and the companion defect analysis gives an exact involutive-compression mode mechanism and explicit mode/no-mode fixtures, but a complete four-site sign-family audit proves that winding alone does not determine the compression signature: wall position relative to the reflection-fixed sites is also required. A constant $|z|$ alone would only reparametrize an assigned Dirac mass; the construction escapes through its oriented data. The gap-closing locus is the geometric collinearity condition, the directed weights and the local connection distinguish z from $|z|$, and a finite quartic pair transfer, together with its exactly unitary Fock-norm-preserving pair kick, has forward two-particle amplitude z and reverse amplitude $\bar{z}$. The same kick admits an exact placement on any embedded four-mode set as an even quartic CAR polynomial. Disjoint sets commute, and a finite unit-phase gate schedule preserves strong CAR support inside the union of its declared sets. No graph-distance cone follows until contiguity and local-layer hypotheses are added. The constant phase is invisible to the one-particle rest spectrum — the rest operators of $3 + 4i$ and 5 are exactly unitarily conjugate — yet it appears as a gauge-invariant relative-phase interference quantity in a supplied two-particle pair-kick sector, with return probability $4/5$ versus 1 at those witnesses. Whether the free walk forces that interaction is open, so this is a witness that the supplied interaction retains the phase, not yet a free-theory escape. The same framework proves

its own boundaries. An exact all-zone determinant factorization, $\det(U^{(4)} \mp \mathbf{1}) = 4P_{0,\pi}$ with explicit polynomials, holds at every momentum and mass angle, and for every massive angle with $0 < |\cos \theta| < 1$ it yields a complete crossing classification: zero- and π -quasienergy modes occur exactly where all three momentum cosines vanish simultaneously. The massless cubic regulator has exactly three non-origin zero-quasienergy corner aliases; the massive body-center step retains ± 1 eigenmodes for every mass angle, so this regulator cannot open a global Floquet gap; no degree-one nearest-neighbor factor supports a stationary amplitude compatible with origin normalization, exact unitarity, and the full Dirac tangent; every momentum-independent onsite coin leaves all three even-parity aliases intact; the positive homogeneous Pluecker action selects only the collapsed zero scale; and exact two-step blocking closes the mass subgroup but not the ordered split family. A pre-registered high-momentum benchmark passes all alias, body-center, Wilson-gap, and negative-control thresholds. Every finite statement above is checked in Lean 4; the novelty is the derived rest operator and its verified dynamical chain, not a new constant-coefficient $1 + 1$ walk. A strict alias-free discrete-time $3 + 1$ regulator, position-space PDE convergence, a local interacting Fock dynamics, and any absolute mass scale remain open.

How to read the status marks. Origin and verification are independent. **CLASSICAL** identifies established mathematics formalized here; **NEW** identifies a new theorem or construction; **REALIZATION** identifies an explicit model. **KERNEL** means the stated finite claim is checked by the pinned Lean kernel; **PROOF** means a complete mathematical proof is given here but its dedicated Lean declaration is pending; **OPEN** marks a named unsolved problem. Physical interpretations are never promoted by a verification mark.

1 Introduction: one invariant, one operator, one walk

The sum of lightlike momenta need not be lightlike. In the spinor representation, each future null momentum is a rank-one positive matrix $\psi\psi^\dagger$. A collection of such momenta remains on the null boundary exactly while its spinors occupy one complex direction. Once two directions open, their exterior area is nonzero and the sum moves into the timelike interior.

This familiar geometry contains a sharper operator statement. The complex area $z = \psi \wedge \phi$ is not merely a number whose squared modulus happens to equal mass squared. It is the off-diagonal entry of a canonical odd Hermitian operator. That operator squares to the invariant momentum determinant, anticommutes with the velocity grading, and therefore supplies the rest term of a finite Dirac Hamiltonian. Its exponential is an exact unitary turn gate. The same object thus controls quadratic cost, first-order rest energy, finite-time rotation, and corner amplitudes.

Synthesis theorem. Let $\psi, \phi \in \mathbb{C}^2$, set $P = \psi\psi^\dagger + \phi\phi^\dagger$, $z = \psi \wedge \phi$, and $\mu = |z|$. Then

$$\det P = \mu^2, \quad B_z = \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad B_z^2 = \mu^2 \mathbf{1}.$$

Writing $\Gamma = \text{diag}(1, -1)$ and $H_z(k) = k\Gamma + B_z$, one has

$$\Gamma B_z + B_z \Gamma = 0, \quad H_z(k)^2 = (k^2 + \det P) \mathbf{1}.$$

Thus the positive rest energy is $\mu = \sqrt{\det P}$, and the gap closes exactly on the collinear null boundary. Multiplying z by a unit phase changes B_z by unitary conjugation, so the spectrum depends only on P while the oriented decomposition retains a path phase. **NEW**

KERNEL

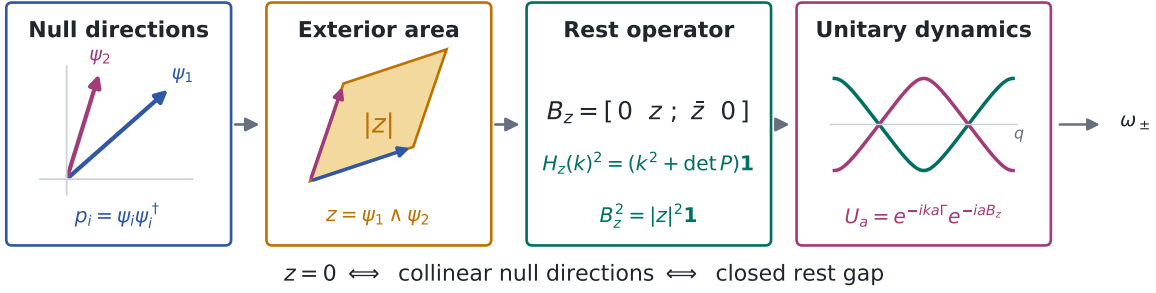


Figure 1: The complete finite construction. The input is a pair of null spinors, not a preassigned mass angle. The complex area fixes both the rest spectrum and the orientation phase of the unitary turn; collinearity is the exact gap-closing boundary.

What changes physically, and what does not. The Pluecker origin fixes more than a dispersion parameter: it fixes the complex directed turn amplitudes, their local chiral connection under a spatially varying field, the exact gap-closing locus, and a finite quartic pair-transfer amplitude that distinguishes z from $\bar{z}$. It does not by itself cure lattice aliases or predict an absolute mass. In fact, within the four-channel separable degree-one architecture, every even-parity zone corner is provably locked to the origin after any onsite Pluecker coin; under the natural positive homogeneous action, every positive configuration has only the collapsed stationary scale. These are falsifiable structural consequences, not interpretive qualifications. NEW [KERNEL](#)

The contributions form two pillars. The first is a derivation: the canonical rest operator and its verified dynamical chain, from spinor pair to walk to histories to uniform continuum control. The second is an obstruction suite: exact all-zone criteria and no-go theorems that fix the boundary of the minimal architecture — including a corollary constraining the newest published doubler-free constructions — and that turn the search for a clean strict $3 + 1$ regulator into a precise algebra problem. Seven contributions organize the paper.

1. **The invariant.** We isolate the exterior-area invariant, its exact massless boundary, and the grading between the first-order gap μ and the quadratic invariant μ^2 . The Cauchy–Binet identity is classical; its complete finite formalization and controls are part of the contribution.
2. **The canonical operator.** We construct B_z directly from the complex Pluecker coordinate, prove the synthesis theorem for arbitrary spinor pairs, and classify every odd Hermitian rest operator as B_w for a unique w . No rational example or separately chosen Dirac mass parameter carries the general statement.
3. **Exact histories.** We derive the exact unitary turn gate and its corner expansion, with finite-spacing corner parameter $\tan(a\mu)$ rather than $a\mu$. A directed-edge path theorem places z and $\bar{z}$ in the same finite history sum and proves equality with the actual complex Pluecker transfer power.
4. **Uniform continuum control.** We prove kernel-checked, uniform bounded-momentum $O(1/n)$ many-step estimates for the $1 + 1$ and ordered $3 + 1$ split-step symbols. The full complex phase is covered by unitary conjugacy and by the direct complex-coin composition, not discarded after taking its modulus.

5. **A finite local $3 + 1$ walk with exact Fourier control.** We construct an exactly norm-preserving finite-torus walk with an explicit two-sided inverse and a strict one-shift-per-axis causal cone, whose ordered symbol is the product controlled by the $3 + 1$ estimate. The finite-to-analytic momentum-sign bridge is exact, an iterated three-torus Plancherel theorem gives the exact N^{-3} wave-packet normalization, and a compact-support multiplier theorem yields momentum-space L^2 convergence at every fixed time. Arbitrary position-dependent Pluecker fields evolve unitarily, and site-dependent chiral rotation induces exact endpoint links.
6. **The obstruction audit.** The same framework proves where it stops, and now does so with an exact all-zone criterion: the determinant factorizations $\det(U^{(4)} \mp \mathbf{1}) = 4P_{0,\pi}$ hold at every real momentum and mass angle, so zero and π modes are exactly the zeros of two explicit polynomials — completely classified for $0 < |\cos \theta| < 1$ as the simultaneous vanishing of all three momentum cosines. The massless cubic regulator has three nonzero zero-quasienergy corner aliases; the massive body-center step retains exact $+1$ and -1 modes for every mass angle, so the regulator is globally ungapped; a degree-one nearest-neighbor factor admits no stationary amplitude compatible with origin normalization, exact unitarity, and the full Dirac tangent; and every momentum-independent onsite coin leaves all three even-parity aliases intact. Exact temporal blocking closes the mass subgroup but generates new ordered words in the full split family, and a homogeneous-action no-go proves that selecting a nonzero $|z|$ requires an additional scale-bearing input.
7. **The many-body seed.** A generic occupation-basis construction satisfies all canonical anticommutation relations, and a four-mode quartic pair transfer has oriented two-particle amplitudes z and $\bar{z}$, exactly distinguished at the fixture $z = 3 + 4i$. The associated normalized pair kick is an explicit complex-linear automorphism, involutive and Fock-inner-product preserving; the Hermitian quartic operator equals the kick on the pair sector and the identity off it, and the kick fixes every one-particle basis state while acting nontrivially on the displayed two-particle state. An interference corollary shows that the supplied interaction sector retains the phase: equal-modulus fields with exactly conjugate one-particle rest operators are separated, inside the supplied pair-kick sector, by a double-kick return probability of $4/5$ versus 1 . This is not yet a second quantization of the spatial walk: the determinant-minor lift now has exact vacuum, one-particle, number, parity, creation-covariance, identity, and composition theorems, including $\Gamma(UV) = \Gamma(U)\Gamma(V)$, and its ordered-minor coefficients obey exact conjugate-transpose compatibility. A one-particle unitary's lifted conjugate transpose is an exact two-sided inverse, and the lift preserves the finite occupation-basis Hermitian inner product. Creation and annihilation covariance are exact. Separately, the pair kick can be placed on arbitrary four-mode sets; disjoint gates commute, and every finite declared schedule has exact support propagation in the even CAR algebra, with a nonzero phase-transfer witness inside the acted set. A bounded-speed cone still requires a graph metric and local-layer theorem. Composing this interaction layer with the spatial free walk, deriving an operator-exponential law, and computing scattering or binding remain open.

The result sits between spinor-helicity kinematics and discrete relativistic quantum dynamics [1, 2]. Cauchy–Binet and the Pauli map supply the classical foundation. Feynman's checkerboard and modern quantum walks show how a massive Dirac propagator can arise from null steps and direction changes [4, 5, 6, 7, 14]. Our contribution is the exact finite map from the oriented spinor area to the odd rest operator and then to a controlled unitary walk, together with a machine-checked dependency boundary.

Construction	Exact local unitary	Exact finite histories	Mass input	Dimension	Continuum control	Machine checked
Feynman checkerboard [4, 5]	not its object	yes	assigned corner weight	1 + 1	propagator limit	no
Dirac QCA [6, 7]	yes	yes	automaton parameter	1 + 1	channel/wave-packet bounds	no
Split-step DCA [8]	yes	not central	tunable coin angle	1 + 1	Dirac limit	no
Weyl/Dirac QCA [10]	yes	model dependent	Weyl coupling	up to 3 + 1	relativistic limit	no
Tetrahedral Dirac walk [11]	yes	not central	assigned $e^{-i\epsilon m\gamma^0}$ coin	3 + 1 tetrahedra	first-order Dirac limit	no
Successive-axis walk [14]	yes	not central	assigned mass parameter	3 + 1	first-order relativistic limit	no
This work	yes	yes	$z = \psi \wedge \phi$	1 + 1; finite 3 + 1 lift	uniform $O(1/n)$ symbols; three-torus L^2	yes

Table 1: Structural comparison, not a priority ranking. The closest QCA works have stronger field-theoretic or higher-dimensional scope. The present distinction is the geometrically generated complex rest operator, the exact kernel dictionary, and their formal verification.

Nearest constructions and the exact novelty claim

For constant z , the update in this paper is unitarily equivalent to the standard 1 + 1 split-step Dirac walk. We do not claim otherwise. What is new is the input map and its naturality: the coin is generated by the complex Pluecker coordinate of null momentum factors, its modulus gives the rest energy, its phase acts by internal conjugacy and survives in oriented turn amplitudes, and the resulting dependency chain is machine checked. Table 1 positions that claim against the closest constructions.

The Pavia programme moreover derives the Weyl and Dirac automata from informational first principles [10]; the derivation offered here is geometric rather than axiomatic, and the two are answers to different questions.

The kinematic layer itself has long roots, and we claim none of it. Weyl–van der Waerden and massive spinor-helicity formalisms routinely decompose a massive momentum into two null spinors, with $\det K = \lambda_1 \lambda_2 = k^2$ appearing explicitly in the eigen-decomposition of the momentum bispinor [3], and the bold-spinor formalism of Arkani-Hamed, Huang, and Huang [2] makes the two-null-spinor description of massive states the organizing principle of amplitude kinematics. Those formalisms use the decomposition as covariant bookkeeping for scattering amplitudes. The present paper starts where that bookkeeping stops: the wedge coordinate is promoted to the off-diagonal entry of a canonical odd Hermitian operator, the operator is classified (Theorem 3.2), exponentiated into an exactly unitary walk, tracked through directed histories and a local connection, and every finite statement in that chain is kernel-checked. The amplitude formalisms neither assert nor need such a dynamical chain; the overlap is the classical 2×2 Cauchy–Binet layer, which we label CLASSICAL throughout.

The reparametrization test

At constant nonzero z , the one-particle spectrum alone cannot distinguish this construction from a Dirac walk whose mass was assigned as $m = |z|$. The claim of added content must therefore survive after that spectrum is held fixed. Table 2 states the test explicitly.

This comparison also fixes the paper’s ceiling. Its centerpiece is a one-particle 1 + 1 walk with a new geometric input and a rigorous finite chain; Section 7 adds a controlled finite 3 + 1 local extension. It is not yet a new QCA classification, an interacting QCA, or a field theory; its exact high-symmetry audit in fact exposes corner aliases and a mass-independent body-center no-gap obstruction that a successor regulator must remove.

Question	Assigned scalar mass	Pluecker-derived input in this paper
Constant dispersion	Same after setting $m = z $	No claimed distinction
Gap-closing locus	External condition $m = 0$	Exact collinearity condition $\psi \wedge \phi = 0$
Directed histories	A real or independently chosen turn coefficient	Opposite orientations carry z and $\bar{z}$ in the same transfer sum
Spatially varying phase	Additional complex-mass field supplied independently	Primitive-spinor phase induces an exact endpoint connection under local chiral conjugacy
Finite two-particle probe	Requires an independent interaction coefficient	The exhibited quartic transfer reuses the same wedge as amplitudes z and $\bar{z}$; equal-modulus fields with conjugate one-particle operators are separated by an exact interference probability (4/5 versus 1) in the supplied pair-kick sector
Absolute scale	Not predicted	Not predicted; the homogeneous-action no-go proves an added scale is required
Ultraviolet lattice spectrum	Regulator dependent	Still regulator dependent; the present walk retains exact aliases

Table 2: What is and is not gained by deriving the mass parameter. The quartic row is an explicit finite extension, not yet a theorem that the free action uniquely forces that interaction.

2 Null-spinor area and the future cone

2.1 From rank-one momenta to area

Let $\psi_1, \dots, \psi_N \in \mathbb{C}^2$ and define

$$M = [\psi_1 \ \cdots \ \psi_N], \quad P = MM^\dagger = \sum_{i=1}^N \psi_i \psi_i^\dagger. \quad (1)$$

Each summand is positive semidefinite and has rank at most one. Under the mostly-minus Pauli map, it is a future null four-momentum. The determinant of P is the Lorentzian square of the total momentum.

Theorem 2.1 (Null-spinor area identity). *For the matrix in (1),*

$$\det P = \left\| \bigwedge^2 M \right\|_{\text{HS}}^2 = \sum_{i < j} |\det [\psi_i \ \psi_j]|^2. \quad (2)$$

Consequently $\det P \geq 0$, and $\det P = 0$ if and only if the occupied columns span a subspace of dimension at most one.

CLASSICAL KERNEL

Proof. Cauchy–Binet applied to $MM^\dagger$ expresses its determinant as the sum of the squared moduli of all 2×2 minors of M . Those minors are exactly the coordinates of $\bigwedge^2 M$. Every summand is nonnegative, so the sum vanishes exactly when every pairwise wedge vanishes, equivalently when $\text{rank } M \leq 1$. $\square$

Define

$$\mu := \left\| \bigwedge^2 M \right\|_{\text{HS}} = \sqrt{\det P}. \quad (3)$$

The quantity μ is the first-order rest energy; μ^2 is the Lorentzian quadratic invariant. The exterior norm makes the geometric word “area” literal rather than metaphorical.

For readers who prefer spatial directions, write the null momenta as $p_i = E_i(1, \mathbf{n}_i)$ with $E_i \geq 0$ and $|\mathbf{n}_i| = 1$. With the same mostly-minus normalization,

$$\mu^2 = 2 \sum_{i < j} E_i E_j (1 - \mathbf{n}_i \cdot \mathbf{n}_j). \quad (4)$$

Mass squared is therefore a positive sum of directional disagreements. It is zero only when every occupied direction agrees; no cancellation between different pairs is available.

2.2 Why the determinant is canonical

The determinant is not selected merely because it gives the desired answer. On the physical space of Hermitian 2×2 matrices it is the unique real quadratic form, up to scale, that vanishes on the full rank-one positive boundary.

Proposition 2.2 (Null-boundary uniqueness). *Let q be a real quadratic form on $\text{Herm}(2)$. If $q(\psi\psi^\dagger) = 0$ for every $\psi \in \mathbb{C}^2$, then $q(P) = c \det P$ for some $c \in \mathbb{R}$.*

CLASSICAL [KERNEL](#)

Proof. Identify $P \in \text{Herm}(2)$ with $(t, \mathbf{x}) \in \mathbb{R}^{1,3}$ so that $\det P = t^2 - |\mathbf{x}|^2$. Write $q(t, \mathbf{x}) = at^2 + 2t\mathbf{b} \cdot \mathbf{x} + \mathbf{x}^T C \mathbf{x}$, with C symmetric. On the future null cone, homogeneity reduces the hypothesis to $q(1, \mathbf{n}) = 0$ for every unit vector $\mathbf{n}$. Comparing $\mathbf{n}$ with $-\mathbf{n}$ gives $\mathbf{b} = 0$. Hence $\mathbf{n}^T C \mathbf{n} = -a$ on the unit sphere, which forces $C = -aI$ by polarization. Thus $q = a(t^2 - |\mathbf{x}|^2) = a \det P$. $\square$

The Lean theorem uses the four real coordinates of a Hermitian matrix and nine explicit rank-one spinor probes to eliminate every quadratic coefficient except one multiple of $pq - r^2 - s^2$. The older real-symmetric slice and a form that fails the full-boundary hypothesis remain as independent controls.

Corollary 2.3 (First-order uniqueness). *Suppose g is nonnegative and homogeneous of degree one on the future cone, and $g(P)^2$ is a quadratic form vanishing on the rank-one boundary. Then $g(P) = c\sqrt{\det P}$ for some $c \geq 0$.*

CLASSICAL [PROOF](#)

This is the logical bridge from a canonical mass-squared invariant to a canonical rest gap. The normalization $c = 1$ fixes units.

3 The Pluecker mass operator

For two spinors, retain the complex orientation data rather than immediately discarding it into an absolute square:

$$z := \psi \wedge \phi = \psi_0 \phi_1 - \psi_1 \phi_0, \quad P := \psi\psi^\dagger + \phi\phi^\dagger. \quad (5)$$

Theorem 2.1 gives $|z|^2 = \det P$. Define

$$B_z := \begin{pmatrix} 0 & z \\ \bar{z} & 0 \end{pmatrix}, \quad \Gamma := \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (6)$$

Theorem 3.1 (Pluecker-derived rest operator). *The matrix B_z is Hermitian and odd for the grading Γ . More precisely,*

$$B_z^\dagger = B_z, \quad \Gamma B_z + B_z \Gamma = 0, \quad B_z^2 = |z|^2 \mathbf{1} = \det(P) \mathbf{1}. \quad (7)$$

For

$$H_z(k) := k\Gamma + B_z, \quad (8)$$

one has

$$\boxed{H_z(k)^2 = (k^2 + \det P) \mathbf{1}}. \quad (9)$$

The rest eigenvalues are $\pm|z|$, and $B_z = 0$ exactly when the two nonzero spinor directions are collinear.

NEW **KERNEL**

Proof. Hermiticity is immediate from the two off-diagonal entries. Direct multiplication gives $B_z^2 = \text{diag}(z\bar{z}, \bar{z}z) = |z|^2 \mathbf{1}$, while the diagonal signs in Γ give the anticommutator zero. Expanding $H_z(k)^2$ leaves $k^2 \mathbf{1} + B_z^2$ because the mixed term vanishes. Finally $B_z = 0$ if and only if $z = 0$, and a two-spinor wedge vanishes exactly on projective collinearity. Every step of this proof is represented by a general Lean declaration, not by evaluation of a fixed example. $\square$

Theorem 3.1 is stronger than assigning the scalar $\det P$ to an otherwise independent Dirac mass slot. The odd operator is constructed from the oriented Pluecker coordinate itself; its square derives the quadratic invariant. The distinction between powers is now fixed:

Layer	Natural quantity
Null-spinor kinematics	$\mu^2 = \det P = z ^2$
Odd mass operator	B_z , with $B_z^2 = \mu^2 \mathbf{1}$
Quadratic costs and Hessians	μ^2
Dirac rest energy	$\mu = z $
Finite unitary turn angle	$a\mu$
Exact corner/stay ratio	$\tan(a\mu)$

3.1 Phase covariance and decomposition independence

If $u \in \mathbb{C}$ has $|u| = 1$, let $W_u = \text{diag}(u, 1)$. Then

$$W_u B_z W_u^\dagger = B_{uz}, \quad W_u W_u^\dagger = \mathbf{1}. \quad (10)$$

Both identities are kernel-checked. Thus the phase of z rotates the internal basis but cannot change the rest spectrum.

The construction is also covariant under the spacetime spin group. If $A \in \text{SL}(2, \mathbb{C})$ acts on both spinors by left multiplication, then

$$(A\psi) \wedge (A\phi) = \det(A)(\psi \wedge \phi) = z, \quad B_{(A\psi) \wedge (A\phi)} = B_z. \quad (11)$$

This is the algebraic $\text{SL}(2, \mathbb{C})$ statement covering proper-orthochronous Lorentz covariance; it is **KERNEL**. General $A \in \text{GL}(2, \mathbb{C})$ acts with the expected determinant weight; that weighted extension is an algebraic **PROOF** corollary, not a separate Lean declaration in the present development.

Theorem 3.2 (Classification of odd Hermitian rest operators). *Let $B \in M_2(\mathbb{C})$ be Hermitian and satisfy $\Gamma B + B\Gamma = 0$. Then there is a unique $w \in \mathbb{C}$ such that*

$$B = B_w = \begin{pmatrix} 0 & w \\ \bar{w} & 0 \end{pmatrix}, \quad B^2 = |w|^2 \mathbf{1}. \quad (12)$$

Consequently the odd Hermitian square roots of $\mu^2 \mathbf{1}$ form one phase circle $w = \mu u$, $|u| = 1$, and for $\mu > 0$ they are all related by the conjugacies in (10). Specifying the oriented Pluecker coordinate z selects one member of that orbit.

NEW [KERNEL](#) for the normal form and square; [PROOF](#) for the phase-orbit corollary.

Proof. Oddness forces both diagonal entries of B to vanish, and Hermiticity forces the lower-left entry to be the complex conjugate of the upper-right entry $w = B_{01}$. The square is then $|w|^2 \mathbf{1}$. If it equals $\mu^2 \mathbf{1}$, then $|w| = \mu$; for $\mu > 0$, write $w = \mu u$ with $|u| = 1$ and apply (10). $\square$

This also controls the nonuniqueness of the null decomposition. If $M = [\psi \ \phi]$ has full rank and $M' = MR$ with $R \in U(2)$, then $M'M'^\dagger = MM^\dagger$ and $\det M' = (\det R) \det M$. Since $|\det R| = 1$, the two mass operators are related by (10). The unitary-equivalence class of B_z depends only on P ; an oriented factorization retains the phase needed by individual turn amplitudes. [CLASSICAL KERNEL](#)

The distinction matters physically. The spectrum is decomposition independent, while an amplitude can remain sensitive to orientation. The construction preserves exactly the information needed by a path sum and forgets exactly the information a rest energy cannot observe.

4 Exact unitary null histories

4.1 Exponentiating the derived rest term

For $\mu = |z| > 0$, set $\hat{B}_z = B_z/\mu$. Since $\hat{B}_z^2 = \mathbf{1}$,

$$e^{-iaB_z} = \cos(a\mu) \mathbf{1} - i \sin(a\mu) \hat{B}_z. \quad (13)$$

At $\mu = 0$ the expression is understood by continuity and equals $\mathbf{1}$. Combining the turn with the conditional null translation gives the momentum symbol

$$U_a(k; z) := e^{-iak\Gamma} e^{-iaB_z}. \quad (14)$$

Each factor is unitary. The two components translate in opposite null directions; only the odd factor changes direction.

The finite formula in (13) is itself closed under composition: writing it as $C_z(a)$, one has

$$C_z(a)C_z(a)^\dagger = \mathbf{1}, \quad C_z(a)C_z(b) = C_z(a+b). \quad (15)$$

The explicit rest eigenvectors, right-unitary decomposition theorem, and this one-parameter group law are all [KERNEL](#).

By phase covariance, $U_a(k; z)$ is unitarily conjugate to the real-mass walk with parameter μ . Its determinant is one and its trace is $2 \cos(ka) \cos(\mu a)$. Therefore an eigenphase $e^{-i\omega a}$ obeys

$$\boxed{\cos(\omega a) = \cos(ka) \cos(\mu a)}. \quad (16)$$

At $k = 0$ the exact eigenphases are $e^{\mp ia\mu}$, so the positive principal-branch quasienergy is μ . Quasienergy is periodic modulo $2\pi/a$; no global unbranched equality is asserted. The real-mass spectral statement and oriented mass-coin group law are [KERNEL](#); the displayed spectral conjugacy is an immediate corollary of the kernel-checked phase covariance.

4.2 The exact corner ratio

Equation (13) gives two kinds of matrix element. A component that keeps its direction contributes $\cos(a\mu)$. A direction change contributes $-i \sin(a\mu)$ times z/μ or $\bar{z}/\mu$, according to its orientation. Hence an n -step history h with $c(h)$ turns has amplitude

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^{n-c(h)} (-i \sin(a\mu))^{c(h)} \Xi_z(h), \quad (17)$$

where $\Phi_k(h)$ is the translation phase and $\Xi_z(h)$ is a product of unit-modulus orientation factors. Whenever $\cos(a\mu) \neq 0$,

$$A_a(h) = e^{i\Phi_k(h)} \cos(a\mu)^n (-i \tan(a\mu))^{c(h)} \Xi_z(h). \quad (18)$$

Proposition 4.1 (Unitary checkerboard correspondence). *After removal of the common factor $\cos(a\mu)^n$, the exactly unitary walk is a corner-weighted checkerboard sum with finite-spacing corner parameter $\tan(a\mu)$ and the orientation phase inherited from z . Under the opposite checkerboard sign convention, the parameter changes sign. In either convention,*

$$\tan(a\mu) = a\mu + O(a^3), \quad (19)$$

but $\tan(a\mu)$ and $a\mu$ are not identical at finite spacing.

CLASSICAL for the coin parametrization; NEW [KERNEL](#) for the exact two-weight recursion scaling and the separate transfer-matrix identification; NEW [KERNEL](#) for the directed complex orientation bookkeeping in the same path sum.

Proof. Insert the two matrix elements from (13) at every time slice. A history with $c(h)$ off-diagonal selections and $n - c(h)$ diagonal selections has exactly the product in (17). Factoring one $\cos(a\mu)$ from every slice gives (18). The Taylor statement is the standard expansion of the tangent. $\square$

The Lean development verifies the complete finite dictionary. Its recursive two-weight kernel satisfies

$$K[s, sw]_n = s^n K[1, w]_n,$$

and specialization to $s = \cos(a\mu)$ and $w = -i \tan(a\mu)$ gives the unitary checkerboard correspondence exactly when $\cos(a\mu) \neq 0$. A quarter-angle nondegenerate witness and the $a\mu = \pi/2$ pure-flip boundary are included. Separately, the development verifies that finite null histories weighted by a corner power equal a closed binomial kernel and satisfy a discrete Dirac recursion, and that the same direction-resolved history sum equals the corresponding transfer-matrix power. The direct Gaussian-rational path-sum declaration uses a real mass and corner weight $+i\epsilon m$ in its stated convention. Independently, the directed-edge theorem assigns every history edge the corresponding entry of the actual complex Pluecker mass coin and proves that the complete oriented history sum equals its transfer-matrix power. Thus Ξ_z in (17) is now packaged directly: opposite turns carry z and $\bar{z}$, with the exact $3 + 4i$ control proving that the two normalized orientation factors are nonzero and unequal.

The tangent appearance is not asserted as novel by itself. Exponential and Euler-angle parametrizations of unitary Dirac-walk coins were related earlier, including explicit tangent formulas in quantum lattice-Boltzmann work [9]. The contribution here is the exact recursive-kernel scaling law composed with a coin whose generator is derived from the complex Pluecker coordinate, together with its machine verification.

5 Full lattice spectrum, symmetry, and doubling audit

Quantum-walk claims cannot be judged only at small momentum. Introduce the dimensionless variables

$$q = ka \in [-\pi, \pi], \quad \theta = \mu a, \quad \Omega = \omega a.$$

The Brillouin zone is periodic in q , and quasienergy is periodic modulo 2π .

Theorem 5.1 (Complete two-band audit). *The two quasienergy bands are*

$$\Omega_{\pm}(q) = \pm \arccos(\cos q \cos \theta) \pmod{2\pi}. \quad (20)$$

They are degenerate only when $q \in \pi\mathbb{Z}$ and $\theta \in \pi\mathbb{Z}$. On the physical principal branch $0 < |\theta| < \pi/2$, the walk is gapped throughout the Brillouin zone: the positive rest energy at $q = 0$ is $|\theta|/a = \mu$, the interband zero-energy gap is 2μ , and the circular quasienergy gap at the zone edge is also $2|\theta|$. At $\theta = 0$ there are two massless cones:

$$(q, \Omega) = (0, 0) \quad \text{and} \quad (q, \Omega) = (\pi, \pi). \quad (21)$$

For a differentiable band away from a degeneracy,

$$v_g = \frac{\partial \omega}{\partial k} = \frac{\sin q \cos \theta}{\sin \Omega}, \quad |v_g| \leq 1, \quad (22)$$

with strict inequality for a massive angle whenever the denominator is nonzero.

REALIZATION [KERNEL](#) for the trace-derived dispersion, velocity deficit inequality, exact massless set $\{0, \pi\}$, and uniform massive discriminant bound; [PROOF](#) for the implicit-differentiation identity in (22) and for packaging these facts into the displayed arccos branches.

Proof. Equation (20) is the principal solution of (16). A two-band degeneracy requires $\sin \Omega = 0$, equivalently $|\cos q \cos \theta| = 1$. Each factor has absolute value at most one, so equality forces $q, \theta \in \pi\mathbb{Z}$. At $q = 0$ one has $\Omega_+ = |\theta|$ on the stated branch; at $q = \pi$ one has $\Omega_+ = \pi - |\theta|$, whose circular distance to the negative band across the quasienergy boundary is $2|\theta|$. Implicit differentiation gives (22). Finally,

$$\sin^2 \Omega - \sin^2 q \cos^2 \theta = 1 - \cos^2 \theta = \sin^2 \theta \geq 0,$$

which proves the speed bound and its strict massive form. $\square$

Doubling and pseudo-doubling verdict. The high-momentum (π, π) cone is a Floquet zone-edge partner of the low-energy cone. It is not a second zero-quasienergy root, but it is a second relativistic band-touching point in the full lattice spectrum. Following the terminology of Gupta and Short [13], it is naturally classified as a pseudo-doubler: a high-quasienergy mode with a Dirac-like local dispersion. The elementary walk therefore does *not* establish a no-doubling or no-pseudo-doubling theorem. The bounded continuum estimate in Section 6 isolates fixed physical momenta as $a \rightarrow 0$, sending $q \simeq \pi$ to cutoff momentum; that decoupling is not the same as removing the finite-lattice partner.

$$\text{Exact bands: } \cos \Omega = \cos q \cos \theta$$

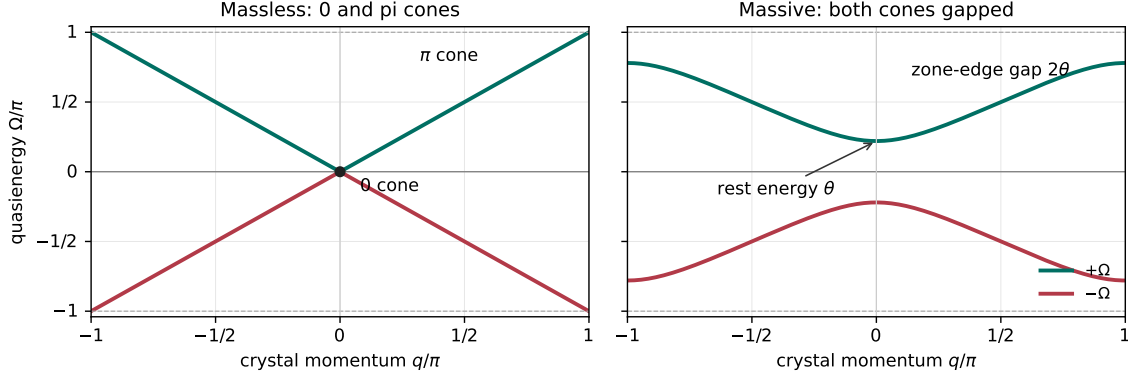


Figure 2: Exact quasienergy bands from $\cos \Omega = \cos q \cos \theta$. The left panel shows the massless cone at the origin and its Floquet partner at quasienergy π . A nonzero Pluecker area opens both gaps (right). The curves evaluate the exact formula; they are not a fit or numerical simulation. Generated by `Scripts/sim/plot_null_spinor_dirac_bands.py`.

Chiral time frame. Let

$$U_s(q, \theta) = e^{-i\theta\sigma_x/2} e^{-iq\sigma_z} e^{-i\theta\sigma_x/2}.$$

It is unitarily equivalent to $U(q, \theta)$ and obeys the exact chiral relation

$$\sigma_y U_s(q, \theta) \sigma_y = U_s(q, \theta)^\dagger. \quad (23)$$

Thus the reciprocal pairing of quasienergies is structural, not an artifact of the trace calculation. Equation (23) is the symmetry statement used here; a finer antiunitary or Floquet topological classification is left to a separate analysis [12].

Robustness. If z is perturbed to $z + \delta z$, then

$$||z + \delta z| - |z|| \leq |\delta z|.$$

Therefore every perturbation with $|\delta z| < |z|$ remains off the collinear gap-closing locus and retains a rest energy at least $|z| - |\delta z|$. The phase may rotate the internal basis, but by (10) it cannot change a constant-coefficient spectrum.

6 Uniform many-step Dirac control

The finite walk should approximate the continuous Dirac flow over a fixed macroscopic time, not merely at one infinitesimal step. For the real representative of the phase-conjugacy class, write

$$U(q, r) = e^{-iq\sigma_z} e^{-ir\sigma_x}, \quad H(k, \mu) = k\sigma_z + \mu\sigma_x, \quad V_t(k, \mu) = e^{-itH(k, \mu)}. \quad (24)$$

Theorem 6.1 (Uniform bounded-momentum product limit). *For $K, M \geq 0$, define*

$$C_{\text{box}}(K, M) = 2K^2 + 2M^2 + KM^2 + K^2M + KM, \quad (25)$$

$$D_{\text{box}}(K, M) = 4C_{\text{box}}(K, M) + 4(K + M)^2 e^{K+M}. \quad (26)$$

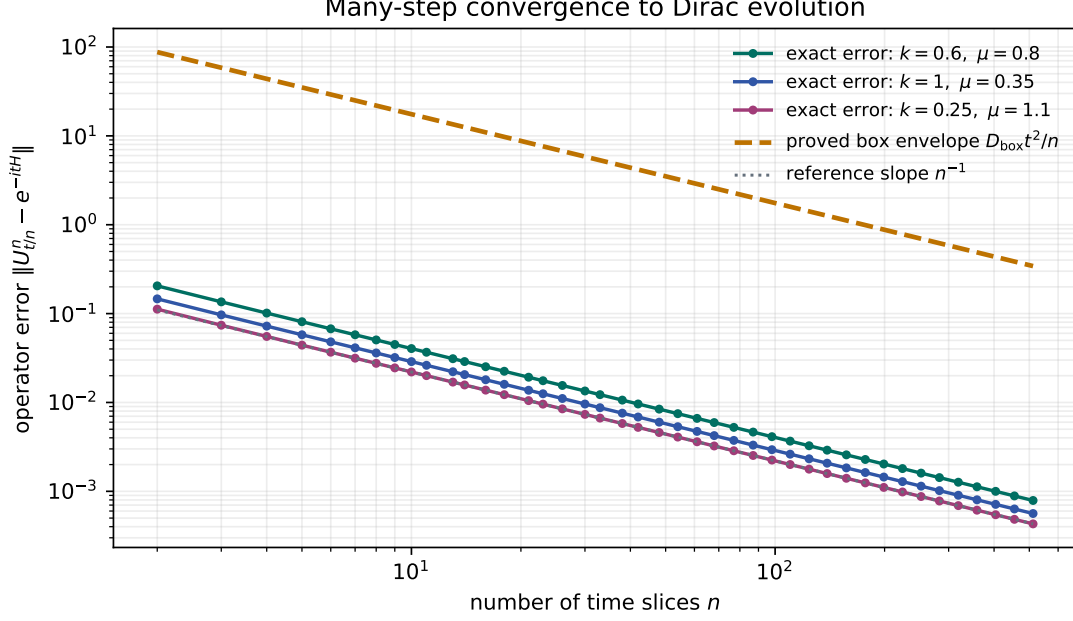


Figure 3: Exact many-step operator errors for three nondegenerate parameter choices. All follow the proved n^{-1} rate and remain below the explicit compact-box envelope. The points evaluate the exact finite matrices; the envelope and convergence statement are kernel checked.

If $|k| \leq K$, $|\mu| \leq M$, $n > 0$, and $|t/n| \leq 1$, then

$$\left\| U\left(k\frac{t}{n}, \mu\frac{t}{n}\right)^n - V_t(k, \mu) \right\| \leq D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (27)$$

The bound is uniform on the displayed parameter box, and its right-hand side tends to zero as $n \rightarrow \infty$.

NEW **KERNEL**

Proof architecture. The one-step walk differs from $\mathbf{1} - i(t/n)H$ by an explicit quadratic remainder. The exact exponential differs from the same first-order matrix by another quadratic remainder, giving $\|U - V_{t/n}\| \leq D(k, \mu)t^2/n^2$. Both matrices are unitary, so the power telescope has no norm-growth factor:

$$\left\| U^n - V_{t/n}^n \right\| \leq n \|U - V_{t/n}\|.$$

The exact short-time flows compose to V_t , producing the $1/n$ rate. Finally $D(k, \mu) \leq D_{\text{box}}(K, M)$ on the parameter box. Lean checks the entrywise remainder estimates, operator-norm conversion, unitarity, telescope, flow composition, box inequality, and limit. $\square$

This proof is a finite-dimensional first-order product-formula argument. Its constant is explicit rather than optimized; sharper general Trotter-error theory provides context for that distinction [15].

For $z = \mu u$ with $|u| = 1$, the diagonal unitary W_u commutes with Γ and conjugates B_μ to B_z . Unitary invariance of the operator norm therefore transfers (27) to the oriented Pluecker walk. This one-line conjugacy corollary is **PROOF** pending a combined declaration.

The theorem proves uniform convergence of the finite momentum symbol in $1 + 1$ dimensions. It is not yet an infinite-volume inverse-Fourier theorem, a position-space PDE convergence theorem, or an interacting limit. Those quantifiers are separate problems, not implications hidden in the word “continuum.”

7 Finite position kernels and a controlled $3 + 1$ extension

The uniform estimates already imply a finite position-space statement. Let $\mathcal{K}$ be a finite momentum set, let $\chi_x(k)$ be phases with $|\chi_x(k)| \leq 1$, and define the synthesized matrix kernel

$$\mathcal{S}_x[A] := \sum_{k \in \mathcal{K}} \chi_x(k) A(k).$$

Applying the triangle inequality to Theorem 6.1 gives the kernel-checked bound

$$\|\mathcal{S}_x[U_n] - \mathcal{S}_x[V_t]\| \leq |\mathcal{K}| D_{\text{box}}(K, M) \frac{t^2}{n}. \quad (28)$$

This controls every finite periodic Fourier kernel. The cardinality factor is the generic price of applying the triangle inequality pointwise. It disappears at the level of normalized L^2 wave packets. For Mathlib’s unnormalized vector-valued transform $\mathcal{F}$ on $\mathbb{Z}/N\mathbb{Z}$, we prove

$$\sum_k \|(\mathcal{F}f)(k)\|^2 = N \sum_x \|f(x)\|^2, \quad \sum_x \|\mathcal{F}^{-1}g(x)\|^2 = \frac{1}{N} \sum_k \|g(k)\|^2. \quad (29)$$

Consequently, if a modewise error obeys $\|\delta(k)\| \leq \varepsilon \|c(k)\|$, then

$$\sum_x \|\mathcal{F}^{-1}\delta(x)\|^2 \leq \frac{\varepsilon^2}{N} \sum_k \|c(k)\|^2. \quad (30)$$

Both identities and the implication are `CLASSICAL KERNEL`; a one-mode packet is included as an exact normalization control. We also prove the operator-valued corollary: a uniform bound $\|A(k) - B(k)\| \leq \varepsilon$ implies (30) for $\delta(k) = (A(k) - B(k))c(k)$. Applying that corollary to the local walk uses the exact plane-wave block stated below. Iterating the transform in the three coordinates gives, for $N \geq 1$, the kernel-checked product-torus identity

$$\sum_{x \in (\mathbb{Z}/N\mathbb{Z})^3} \|\mathcal{F}_3^{-1}g(x)\|^2 = \frac{1}{N^3} \sum_{k \in (\mathbb{Z}/N\mathbb{Z})^3} \|g(k)\|^2. \quad (31)$$

Hence, again for $N \geq 1$, a uniform modewise operator error ε gives the exact wave-packet coefficient ε^2/N^3 , with no cardinality loss. The momentum-sign bridge is also kernel-checked: for the positive-exponential convention the finite character block equals the analytic block at $k_j\varepsilon = -2\pi n_j/L$, including an exact quarter-zone $-i$ sign control.

There is also a countable successor to the finite Plancherel statement. If $e_n(k)$ is a countable family of mode errors such that $e_n(k) \rightarrow 0$ for every k and $\|e_n(k)\|^2 \leq b(k)$ for one summable envelope b , then

$$\sum_k \|e_n(k)\|^2 \longrightarrow 0. \quad (32)$$

This is a direct vector-valued Tannery specialization, proved in Lean with a nonzero one-mode control. It isolates the remaining physical obligation: construct the walk-specific square-summable

envelope and identify the limiting multiplier with the Dirac PDE propagator under a continuum Fourier isometry.

The corresponding measure-theoretic multiplier step is also closed. If $f \in L^2$ and pointwise errors satisfy $\|e_n(k)\| \leq \varepsilon_n \|f(k)\|$ almost everywhere, where $0 \leq \varepsilon_n$ and $\varepsilon_n \rightarrow 0$, then

$$\|e_n\|_{L^2} \rightarrow 0. \quad (33)$$

The proof is a kernel-checked ‘eLpNorm’ monotonicity-and-squeeze argument. The walk-specific compact-support application is now closed as well. For any fixed $t \in \mathbb{R}$, suppose $f \in L^2$ and, almost everywhere on the support of f , $|k_x|, |k_y|, |k_z| \leq K$ and $|z| \leq M$, with $K, M \geq 0$. Applying the complex split symbol for $n + 1$ steps gives an error multiplier E_n satisfying

$$\|E_n f\|_{L^2} \rightarrow 0, \quad \|E_n(k)f(k)\| \leq \frac{D_{K,M} t^2}{n+1} \|f(k)\|. \quad (34)$$

The displayed pointwise envelope holds for all sufficiently large n ; the finite prefix is irrelevant to the sequence limit. This walk-specific composition is `NEW KERNEL`. What remains is the lattice/continuum Fourier-isometry and identification of the limiting multiplier with the position-space Dirac PDE, not compact-support multiplier convergence.

The same architecture extends to four internal components. The successive-axis factorization itself is the standard construction used by Mlodinow and Brun [14]; what is new here is the Pluecker-derived complex coin, the exact finite-torus implementation with a two-sided inverse and strict causal cone, and the kernel-checked all-zone audit that follows. Let Hermitian matrices $\alpha_1, \alpha_2, \alpha_3, \beta \in M_4(\mathbb{C})$ satisfy

$$\alpha_i^2 = \beta^2 = \mathbf{1}, \quad \{\alpha_i, \alpha_j\} = 0 \ (i \neq j), \quad \{\alpha_i, \beta\} = 0,$$

and set

$$H_4(\mathbf{k}, m) = \sum_{j=1}^3 k_j \alpha_j + m \beta.$$

The explicit matrices used in the artifact obey

$$H_4(\mathbf{k}, m)^2 = (|\mathbf{k}|^2 + m^2) \mathbf{1}. \quad (35)$$

For a substep ε , define

$$U_\varepsilon^{(4)}(\mathbf{k}, m) = e^{-i\varepsilon k_x \alpha_1} e^{-i\varepsilon k_y \alpha_2} e^{-i\varepsilon k_z \alpha_3} e^{-i\varepsilon m \beta}. \quad (36)$$

The order in (36) is not cosmetic: the spatial generators do not commute, and the artifact contains that noncommutation as a negative control against a false second-order claim.

Theorem 7.1 (Finite local $3 + 1$ walk and compact rate). *On a finite periodic three-dimensional position register, there is an explicit successive-axis update built from one-site conditional shifts and the mass coin in (36). It preserves the finite inner product exactly, has an explicit two-sided inverse, and reads only sites reached by one conditional source shift along each axis in one cycle. Its product plane-wave sectors are invariant, and on each sector the update is exactly the ordered axis-0/axis-1/axis-2/mass matrix block built from finite roots of unity. For the positive-exponential plane-wave convention, that block is (36) when*

$$k_j \varepsilon = -\frac{2\pi n_j}{L}. \quad (37)$$

If $|k_x|, |k_y|, |k_z| \leq K$, $|m| \leq M$, $n > 0$, and $|t/n| \leq 1$, then

$$\left\| \left(U_{t/n}^{(4)}(\mathbf{k}, m) \right)^n - e^{-itH_4(\mathbf{k}, m)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (38)$$

Every finite Fourier synthesis obeys the same bound multiplied by its number of modes, while the normalized product-torus L^2 wave-packet error obeys the squared bound divided by N^3 . Moreover, let $B_4(z)$ be the phase-retaining rest operator and let $U_\varepsilon^{(4)}(\mathbf{k}, z)$ be the chiral conjugate of the displayed step at $m = |z|$. Its exact comparison flow is generated by

$$H_4(\mathbf{k}, z) = \sum_{j=1}^3 k_j \alpha_j + B_4(z), \quad H_4(\mathbf{k}, z)^2 = (|\mathbf{k}|^2 + |z|^2) \mathbf{1},$$

and whenever $|z| \leq M$ it satisfies the same bound

$$\left\| \left(U_{t/n}^{(4)}(\mathbf{k}, z) \right)^n - e^{-itH_4(\mathbf{k}, z)} \right\| \leq 16(3K + M)^2 e^{3K+M} \frac{t^2}{n}. \quad (39)$$

REALIZATION KERNEL for exact local norm preservation, Clifford square, finite-character plane-wave block, compact rate, finite synthesis bound, and the trigonometric/sign conversion (37) to the analytic symbol; **NEW KERNEL** for the complex-phase transfer (39) and the iterated three-torus normalization (31).

Proof architecture. Each α_j has eigenvalues ± 1 . An explicit unitary eigenbasis turns the corresponding diagonal one-site conditional shift into the physical Clifford factor; composing those shifts with the mass coin proves exact norm preservation. Direct substitution into every product plane wave extracts one unit-modulus character per conditional shift and proves the exact ordered finite block without an asymptotic argument. The closed form $e^{-iq\alpha_j} = \cos q \mathbf{1} - i \sin q \alpha_j$ identifies the ordered analytic matrix symbol after the sign convention in (37). A four-factor quadratic remainder estimate, exact unitarity, and the unitary power telescope then give (38). Applying the finite synthesis theorem gives its position-kernel corollary. $\square$

For $z \neq 0$, the phase transfer is exact. If W_z is the chiral unitary at the polar angle of z , then

$$W_z H_4(\mathbf{k}, |z|) W_z^\dagger = H_4(\mathbf{k}, z), \quad W_z e^{-itH_4(\mathbf{k}, |z|)} W_z^\dagger = e^{-itH_4(\mathbf{k}, z)}.$$

The operator norm cannot increase under this conjugation, so the real-mass estimate transfers without changing its constant.

The 3 + 1 high-symmetry verdict

The continuum tangent is exactly isotropic: $H_4(\mathbf{k}, 0)^2 = |\mathbf{k}|^2 \mathbf{1}$. The finite cubic walk is not globally doubler-free. At the eight corners $(q_x, q_y, q_z) \in \{0, \pi\}^3$, its massless symbol is

$$U^{(4)}(q_x, q_y, q_z, 0) = (-1)^r \mathbf{1}, \quad r = \#\{j : q_j = \pi\}. \quad (40)$$

Thus $(\pi, \pi, 0)$, $(\pi, 0, \pi)$, and $(0, \pi, \pi)$ are distinct momenta with exactly the same zero-quasienergy symbol as the origin. Equation (40), all three distinctness witnesses, and radial isotropy of the continuum square are **REALIZATION KERNEL**. This is an actual finite-regulator obstruction, not a caveat inferred from a plot.

The massive walk has a stronger interior obstruction. At body-center momentum $\mathbf{q}_B = (\pi/2, \pi/2, \pi/2)$, let $u_\theta = \cos \theta + i \sin \theta$ and

$$v_+(\theta) = (-u_\theta, 0, 1, 0)^T, \quad v_-(\theta) = (u_\theta, 0, 1, 0)^T.$$

For every real mass angle θ , the exact finite step obeys

$$U^{(4)}(\mathbf{q}_B, \theta)v_+(\theta) = v_+(\theta), \quad U^{(4)}(\mathbf{q}_B, \theta)v_-(\theta) = -v_-(\theta), \quad v_\pm(\theta) \neq 0. \quad (41)$$

Equivalently, the body-center step is a non-scalar involution for all θ . It therefore retains exact quasienergies 0 and π even after the mass coin is turned on: the present cubic regulator admits no uniform Floquet gap (a body-center crossing at 0 and π persists for every mass angle). The closed body-center matrix, involution identity, and both explicit eigenmodes are NO-GO [KERNEL](#).

The obstruction is also visible in an exact all-zone criterion. Write $x = \cos q_x$, $y = \cos q_y$, $w = \cos q_z$, and $c = \cos \theta$, and define

$$S = 4c^2x^2y^2w^2 - 2c^2x^2y^2 - 2c^2x^2w^2 + c^2x^2 - 2c^2y^2w^2 + c^2y^2 + c^2w^2 - 2x^2y^2w^2 + x^2y^2 + x^2w^2 + y^2w^2, \quad (42)$$

$$P_0 = S - 2cxyw, \quad P_\pi = S + 2cxyw. \quad (43)$$

For the live ordered successive-axis symbol, the finite determinant expansion is now exact at every real momentum and mass angle:

$$\det(U^{(4)} - \mathbf{1}) = 4P_0, \quad \det(U^{(4)} + \mathbf{1}) = 4P_\pi. \quad (44)$$

Thus $P_0 = 0$ and $P_\pi = 0$ are respectively necessary and sufficient for zero- and π -quasienergy modes. At body center $x = y = w = 0$, both polynomials vanish for every θ , recovering (41). The two determinant identities, their explicit 4×4 expansion, and the definitional bridge to the walk used in the continuum estimate are NO-GO [KERNEL](#).

The zero sets are now classified on the principal massive branch. If $0 < |c| < 1$ and $|x|, |y|, |w| \leq 1$, then

$$P_0 = 0 \iff x = y = w = 0 \iff P_\pi = 0. \quad (45)$$

The proof uses the exact decomposition

$$P_0 = (cx - yw)^2 + (y^2 + w^2 - 2y^2w^2)(c^2 + x^2(1 - 2c^2)), \quad (46)$$

whose second factor is strictly positive and whose first prefactor is nonnegative on this branch. Equality then forces all three momentum cosines to vanish; the P_π statement follows by $c \mapsto -c$. Thus the simultaneous body-center cosine locus is the complete zero- and π -quasienergy crossing set for every principal massive angle, not merely a sampled obstruction. The hypotheses are load-bearing: at $c = 0$, additional zeros occur, including $(x, y, w) = (1, 0, 0)$. The classification and this boundary control are NO-GO [KERNEL](#). They classify the failure of the present regulator; they do not make it alias-free.

A degree-one stationary-amplitude no-go. The most economical proposed repair is to add a stay-put channel to each nearest-neighbor factor. The exact ansatz is

$$F(q) = e^{iq}A + B + e^{-iq}C. \quad (47)$$

Let M be any Hermitian involution. If $F(q)$ is unitary for every real q , $F(0) = \mathbf{1}$, and $F'(0) = -iM$, then

$$B = 0. \quad (48)$$

The proof evaluates the exact unitarity residual at $q = \pm\pi/2$; adding the two resulting matrix identities forces $B^\dagger B = 0$. Specializing $M = \alpha_j$ rules out a nonzero stationary amplitude in a degree-one factor with the full Dirac tangent, even before zone-edge separation is imposed. All three specializations to the live $3 + 1$ generators are separately kernel-checked. This is NO-GO [KERNEL](#). A nonvacuity control shows that a partial-channel Hermitian tangent with a stationary kernel does admit an exactly unitary degree-one witness with $B \neq 0$ and a non-scalar zone edge. That witness does not retain the full Dirac tangent. Thus stationary amplitudes remain viable only after a genuine change of architecture, such as a larger internal cell, longer Laurent range, multistep factorization, or a non-involutory tangent.

The last route is not hypothetical. Gupta and Short [13] construct doubler- and pseudo-doubler-free nearest-neighbor one-step unitary walks, in $1 + 1$ and $3 + 1$ dimensions, whose enabling resource is exactly a nonzero stay-put amplitude $\gamma_0 = \mathbf{1} - \gamma_+ - \gamma_-$ built from tilted, non-orthogonal velocity projectors (their Eqs. (29)–(30) and (37)). Their step is exactly of the degree-one Laurent form (47), unitary at every momentum and normalized at the origin, with stationary block γ_0 . The no-go (48) therefore yields a corollary about their family: its tangent generator, though automatically Hermitian for a unitary origin-normalized step, cannot be an involution ($M^2 \neq \mathbf{1}$). The stay-put freedom that removes the doublers is purchased exactly by relinquishing the involutory unit-speed Dirac tangent, and our partial-channel escape witness realizes the same mechanism in kernel-checked form. Their $3 + 1$ family moreover retains extraneous Weyl-like low-energy solutions at isolated momenta $\pm \mathbf{q}(\theta)$, which the authors suggest a more general higher-dimensional construction might remove (their Appendix F), so a strictly local, exactly unitary, fully alias-free $3 + 1$ walk with the complete involutory Dirac tangent remains open in both programs.

The obstruction is stronger for a separable three-axis product. Once $B_x = B_y = B_z = 0$, origin normalization forces each factor to obey $F_j(\pi) = -\mathbf{1}$. Therefore, for any momentum-independent onsite coin Q , including the complex Pluecker mass coin,

$$U_Q(0, 0, 0) = U_Q(\pi, \pi, 0) = U_Q(\pi, 0, \pi) = U_Q(0, \pi, \pi). \quad (49)$$

This all-coins alias theorem is NO-GO [KERNEL](#). It proves that changing the onsite mass data cannot repair the cubic partners inside the four-channel, range-one, single-factor-per-axis architecture. Extra range, coupled substeps, or enlarged/ancillary cells are not optional embellishments; at least one is structurally required.

An attempted commutator–Wilson repair sharpens that requirement. Consider the exactly unitary one-axis gate

$$W_r(q) = F_\beta(rq)F_{\gamma_5}(rq)F_\beta(-rq)F_{\gamma_5}(-rq), \quad F_A(s) = \cos s \mathbf{1} - i \sin s A. \quad (50)$$

Its linear term cancels, so at the Bloch-matrix level it appears to supply a second-order Wilson-like correction without disturbing the Dirac tangent. However, in the fixed one-cell ansatz strict finite Laurent range requires the frequency r to be an integer. For every integer r the exact zone-edge identity is

$$W_r(\pi) = \mathbf{1}, \quad (51)$$

and all three even-corner aliases in (49) remain exactly equal to the origin. Conversely, the noninteger trigonometric values that lift selected corners are not 2π -periodic finite Laurent polynomials in e^{iq}

and therefore do not define the claimed strict local one-cell QCA. This family-scoped locality/dealiasing incompatibility, including the range-one control, is NO-GO **KERNEL**. It does not exclude a larger cell or a genuinely different finite product; it prevents a nonlocal Bloch control from being mislabeled as that successor.

There is already a constructive Hamiltonian-level route around this no-go. Using the same live Clifford generators, define

$$M_W(\mathbf{q}) = m + r \sum_{j=1}^3 (1 - \cos q_j), \quad (52)$$

$$H_W(\mathbf{q}) = \sum_{j=1}^3 \sin q_j \alpha_j + M_W(\mathbf{q}) \beta. \quad (53)$$

The kernel-checked Wilson identity is

$$H_W(\mathbf{q})^2 = \left(\sum_{j=1}^3 \sin^2 q_j + M_W(\mathbf{q})^2 \right) \mathbf{1}. \quad (54)$$

For $m > 0$ and $r \geq 0$, the scalar coefficient is bounded below by m^2 . For $m = 0$ and $r > 0$, it vanishes if and only if $\cos q_1 = \cos q_2 = \cos q_3 = 1$; hence all non-origin cubic corners are lifted. For example, the $(\pi, 0, 0)$ energy squared is exactly $4r^2$. This closes the spectral design problem at the nearest-neighbor Hamiltonian level while preserving the Dirac tangent. It does *not* yet close the stricter QCA problem: $\exp(-itH_W)$ is exactly unitary, but its finite-time exponential is not asserted to be a finite Laurent polynomial or a strictly finite-range one-step update.

Exact temporal blocking and nonclosure. Define the two-step blocking map by $\mathcal{B}(U) = U^2$. The mass-only Pluecker subgroup closes exactly,

$$\mathcal{B}(C_z(\varepsilon)) = C_z(2\varepsilon),$$

but the ordered spatial split family does not close by doubling its scalar step. At the exact x - y quarter-turn witness,

$$\mathcal{B}(U_{xy}(\pi/2)) = -\mathbf{1}, \quad U_{xy}(\pi) = \mathbf{1}. \quad (55)$$

The smallest multiplicatively closed enlargement is the submonoid generated by all bare split steps. These statements are NEW **KERNEL**: temporal blocking generates ordered word/coupling data rather than merely a renormalized copy of the original scalar parameters. Spatial decimation is a separate, stronger RG problem.

A proved boundary on scale selection. For the finite positive Pluecker action developed here, simultaneous scaling of both primitive spinors by t gives the exact law

$$S(t\psi, t\phi; x) = t^4 S(\psi, \phi; x). \quad (56)$$

Whenever the unscaled action is positive, its radial equation $4t^3 S(\psi, \phi; x) = 0$ is equivalent to $t = 0$. The orthogonal-spinor, positive-quartet control has $S = 1/2$, so the obstruction is nonvacuous. This is NO-GO **KERNEL**: the natural positive homogeneous action derives the curvature associated with a supplied z but cannot spontaneously select a nonzero $|z|$. A competing homogeneity, constraint, ensemble scale, or other dimensionful input must be displayed. The exact positive control $V_c(t) = (t^2 - c)^2$ has stationary points precisely at $t = 0$ or $t^2 = c$; for supplied $c = 1$, the nonzero points $t = \pm 1$ are global minima. Selection is therefore possible, but its magnitude is exactly the new input. This control is also REALIZATION **KERNEL**.

High-symmetry audit of the finite 3+1 split walk

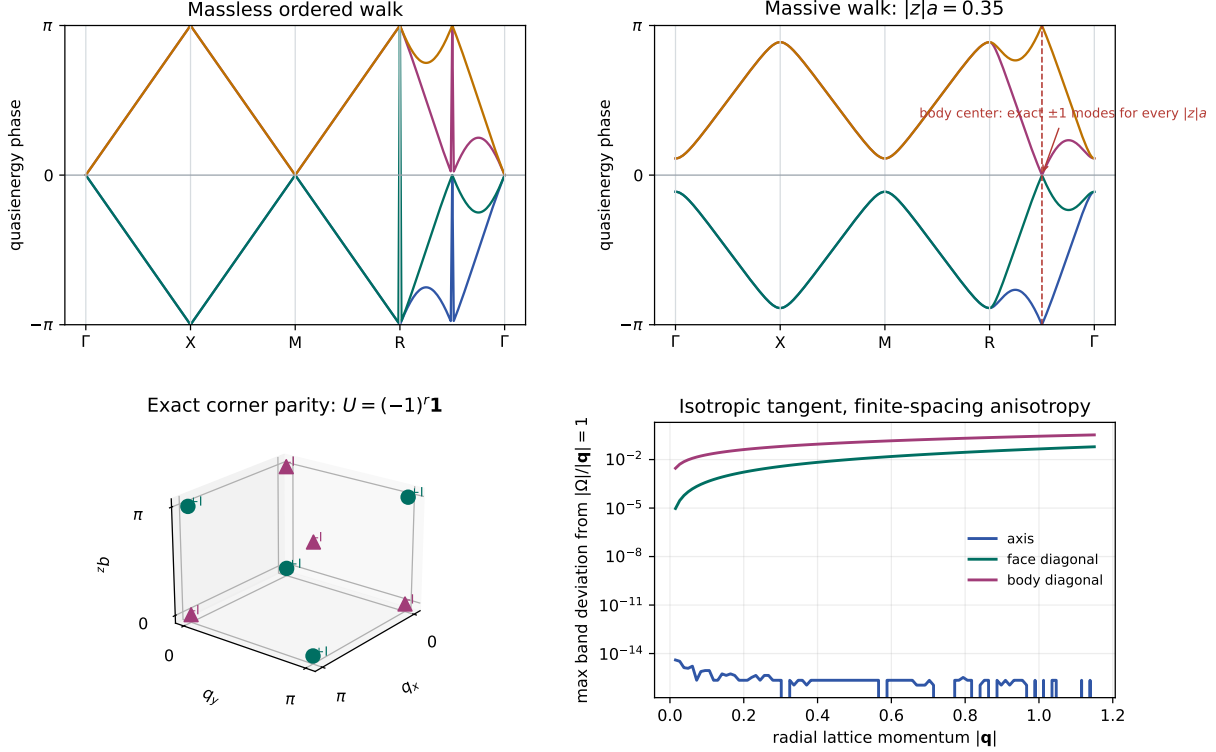


Figure 4: High-symmetry audit of the ordered 3 + 1 walk. The small-momentum tangent is radially isotropic, but the exact massless lattice symbol has four zero-quasienergy corners: the origin and three even-parity aliases. The marked body-center crossing in the massive panel persists for every mass angle by Eq. (41). Curves evaluate the exact finite matrices; the corner classification and body-center eigenmodes are kernel checked.

Pre-registered high-momentum benchmark. Before evaluation we fixed $z = 3 + 4i$, $\varepsilon = 0.2$, Wilson $r = 1$, and a 10^{-10} kill threshold. The oracle then found exactly three non-origin even-corner aliases (maximum residual 2.30×10^{-16}), both body-center ± 1 modes (residuals below 2.24×10^{-16}), no non-origin Wilson zero, and minimum Wilson corner gap 2. With $r = 0$, all eight corners became gapless as required by the negative control. No parameter was fitted. This numerical result validates the implementation of the theorem chain; the Lean identities, not the benchmark, carry the proof.

Comparison with the tetrahedral walk. Nzongani et al. [11] place four amplitudes on tetrahedral facets, compose facet/strict/weakly local shifts with directional coins, and insert the assigned coin $e^{-i\varepsilon m \gamma^0}$; their expansion recovers the 3 + 1 Dirac equation to first order. Our register and finite update are different: the present construction is a cubic successive-axis product with a Pluecker-derived complex mass coin and an explicit many-step rate. A constant onsite basis change cannot decide away this difference: unitary conjugacy preserves equality of symbols at distinct momenta, so every constant onsite conjugate of our walk inherits the three aliases above and the body-center $\{+1, -1\}$ spectrum. Any equivalence with a tetrahedral update must therefore reproduce those obstructions or use a non-onsite blocking, changed unit cell, or coarse-graining map. A full spectral audit of the tetrahedral symbol is needed before choosing among those possibilities.

8 Formal verification and negative controls

The formal development uses Lean 4.28.0 and Mathlib [16, 17]. Its spinor and physics conventions were also cross-checked against the public PhysLean corpus [18]. Formalized mathematical physics at substantial scale is becoming practical: the free bosonic quantum field theory in four-dimensional Euclidean spacetime, satisfying the Glimm–Jaffe axioms, was recently constructed end to end in Lean 4 [19]. The present artifact is complementary in scope: exact finite discrete dynamics, with the physical input — the rest operator itself — derived inside the formalization rather than axiomatized. The trusted Cauchy–Binet identity lives in the project core; the newer operator and walk results remain in the draft namespace while their publication interfaces are reviewed. Each headline declaration below has a build-enforced assumption footprint contained in `{propext, Classical.choice, Quot.sound}`.

Declaration	Status	Mathematical payload
two_edge_plucker_mass_identity	CLASSICAL KERNEL	$\det P = \psi \wedge \phi ^2$
fin_bundle_plucker_mass_identity	CLASSICAL KERNEL	finite-family Cauchy–Binet area sum
hermitian_determinant_uniqueness	CLASSICAL KERNEL	determinant is the unique quadratic null-boundary form up to scale
pair_diracSymbol_sq	NEW KERNEL	$H_z(k)^2 = (k^2 + \det P)\mathbf{1}$ for arbitrary pairs
massOperator_eq_zero_iff_collinear	NEW KERNEL	exact gap-closing locus
phase_covariance	NEW KERNEL	oriented phase acts by conjugacy
massOperator_sl2_invariant	NEW KERNEL	invariance under the defining $\mathrm{SL}(2, \mathbb{C})$ action
oddHermitian_eq_massOperator	NEW KERNEL	complete normal form for odd Hermitian rest operators
massCoin_unitary_group	NEW KERNEL	exact unitary rest evolution and group law
unitary_kernel_eq_scaled_checkerboard	REALIZATION KERNEL	exact $\cos^n$ scaling of the two-weight recursion; operator/history identification is supplied by separate transfer-matrix theorems
exact_quantum_walk_dispersion	REALIZATION KERNEL	concrete trace identity, trace-to-quasienergy bridge, unitarity, determinant one, and rest eigenphases
massive_implies_subluminal	REALIZATION KERNEL	strict full-zone velocity-deficit inequality for massive angle; derivative identity is PROOF
massless_gapless_classification	REALIZATION KERNEL	exact $\{0, \pi\}$ massless band-touching set
pathAmplitude_eq_corner_power	REALIZATION KERNEL	exact real-mass Gaussian-rational corner-power weight; complex orientation is carried by phase conjugacy
complexPlueckerPathSum_eq_transfer_power	NEW KERNEL	direct directed-edge history sum using the actual complex coin entries z and $\bar{z}$
fixed_time_many_step_bound	NEW KERNEL	fixed-parameter $D(k, \mu)t^2/n$ estimate
fixed_time_many_step_bound_on_norm	NEW KERNEL	uniform bounded-parameter estimate
localStep_preserves_norm	REALIZATION KERNEL	exact finite-torus $3 + 1$ norm preservation
inverseSuccessiveWalk_after_successiveWalk_after_inverse	REALIZATION KERNEL	explicit two-sided inverse of the complete local cycle
successiveWalk_zero_of_causal_source_zero	REALIZATION KERNEL	strict one-shift-per-axis causal cone
Finite3Plus1FourierBridge.localStep_modes	REALIZATION KERNEL	exact finite-character block on every product plane wave
finiteLocalSymbol_eq_analytic	REALIZATION KERNEL	exact finite-to-analytic momentum-sign conversion
quarter_zone_sign_control	REALIZATION KERNEL	nondegenerate $-i$ convention fixture
Pluecker3Plus1ComplexMass.H4_sq	NEW KERNEL	phase-retaining $3 + 1$ mass gives $H_4^2 = (\mathbf{k} ^2 + z ^2)\mathbf{1}$
complex_phase_covariance	NEW KERNEL	chiral covariance of the complex four-component rest operator
massCoin4_unitary_group	NEW KERNEL	derived complex mass coin is unitary with exact group law
variableComplexLocalStep_preserves_norm	NEW KERNEL	arbitrary position-dependent Pluecker profiles, including zero sites, evolve unitarily
variableComplexLocalStep_zero_of_causal_source_zero	NEW KERNEL	strict one-shift-per-axis causal cone for the inhomogeneous complex-Pluecker walk
local_phase_connection_covariance	NEW KERNEL	site-dependent phase induces exact endpoint links and rotates $z(x)$ pointwise
gaugedCliffordAxisShift_constant	NEW KERNEL	constant phase produces no endpoint link
global_real_lift_forces_zero_winding	NEW KERNEL	a global real phase lift has zero winding; nonzero index needs branch/transition data
threeLinkUnitWinding_not_global	NEW KERNEL	an explicit periodic link field has winding one and cannot be integrated to a global real vertex phase
totalTurning_int_multiple	NEW KERNEL	the link increments derived from any nowhere-zero Pluecker field (principal arguments of neighbor ratios) sum to an exact 2π -integer winding
windingOneField_totalTurning	NEW KERNEL	an explicit four-site field has derived total turning ex

The controls are part of the result rather than decorative examples.

Claim	Positive/nondegenerate control	Failure or boundary control
Area creates a gap	standard basis pair gives $z = 1$ and $B_z^2 = 1$	collinear pair $(1, 2), (3, 6)$ gives $z = 0$ and $B_z = 0$
Full null boundary matters	determinant vanishes on every rank-one edge	the form ac vanishes on coordinate rays but fails on the diagonal ray
Mass enters at turns	a three-step history is nonzero and mass-dependent	at zero mass every positive-corner history vanishes
Many-step convergence	rational $(k, \mu) = (3/5, 4/5)$ satisfies the bound	reversing the mass phase converges to the wrong Hamiltonian numerically
3 + 1 factor composition	$(1, 2, 2, 3)$ gives $H_4^2 = 181$ and nonzero norm	$\alpha_1\alpha_2 \neq \alpha_2\alpha_1$ forbids order erasure and naive second order

The principal verification commands are

```
lake env lean PhysicsSM/Draft/NullEdge/PluckerMassOperator.lean
lake env lean PhysicsSM/Draft/NullEdge/PluckerMassDynamics.lean
lake env lean PhysicsSM/Draft/NullEdge/CornerWalkEquivalence.lean
lake env lean PhysicsSM/Draft/NullEdge/FermionDoublingAudit.lean
lake env lean PhysicsSM/Draft/NullEdge/ExactCheckerboardPathSum.lean
lake build PhysicsSM.Draft.NullEdge.ExactQuantumWalkDispersion
lake build PhysicsSM.Draft.NullEdge.FixedMomentumManyStepContinuum
lake build PhysicsSM.Draft.NullEdge.BoundedMomentumManyStepContinuum
lake build PhysicsSM.Draft.NullEdge.Compact3Plus1DiracRate
lake env lean PhysicsSM/Draft/NullEdge/Local3Plus1RateBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteWalkPositionConvergence.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteZModPlancherel.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1FourierBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1AnalyticSignBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/Pluecker3Plus1ComplexMass.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerLocalWalk.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerRateTransfer.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteTorus3Plancherel.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteTorus3WalkWavepacket.lean
lake env lean PhysicsSM/Draft/NullEdge/Finite3Plus1BrillouinAudit.lean
lake env lean PhysicsSM/Draft/NullEdge/StationaryAmplitudeNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/StationaryAmplitudeLiveAxisNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteWalkOnsiteEquivalenceObstruction.lean
lake env lean PhysicsSM/Draft/NullEdge/LocalQCAProperties.lean
lake env lean PhysicsSM/Draft/NullEdge/CountableL2WavepacketConvergence.lean
lake env lean PhysicsSM/Draft/NullEdge/ContinuumL2MultiplierBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/CompactSupportL2WalkBridge.lean
lake env lean PhysicsSM/Draft/NullEdge/VariablePlueckerLocalWalk.lean
lake env lean PhysicsSM/Draft/NullEdge/VariablePlueckerPhaseConnection.lean
lake env lean PhysicsSM/Draft/NullEdge/ComplexPlueckerCheckerboardPathSum.lean
lake env lean PhysicsSM/Draft/NullEdge/GlobalPhaseWindingNoGo.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerWindingDerived.lean
lake env lean PhysicsSM/Draft/NullEdge/StrictQCAMinimalArchitecture.lean
lake env lean PhysicsSM/Draft/NullEdge/CommutatorWilsonStrictnessKill.lean
```

```
lake env lean PhysicsSM/Draft/NullEdge/TemporalBlockingRG.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochSplitPlus.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochSplitMinus.lean
lake env lean PhysicsSM/Draft/NullEdge/FullBlochZeroClassification.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteCARFockBasic.lean
lake env lean PhysicsSM/Draft/NullEdge/FiniteCARSecondQuantization.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerQuarticInteraction.lean
lake env lean PhysicsSM/Draft/NullEdge/PlueckerPhaseObservable.lean
lake env lean PhysicsSM/Draft/NullEdge/ChangingLatticePDECore.lean
lake env lean PhysicsSM/Draft/NullEdge/Carrier/PluckerScaleSelectionNoGo.lean
lake build PhysicsSM.Draft.NullEdge.OvernightTheoryAxiomGuard
python Scripts/sim/null_edge_regulator_benchmark.py
python Scripts/sim/plot_null_spinor_dirac_bands.py
python Scripts/sim/plot_null_edge_paper_figures.py
python Scripts/publication/verify_null_edge_paper_a.py
```

The cited modules contain no proof handoff markers or compiled-evaluator proofs. The working tree is not yet an archival release. A clean commit, source archive, theorem ledger, and DOI remain submission requirements.

What mechanization changed

The formal artifact is scientifically useful only where it improves the mathematics. In this development it forced ten corrections that are visible in the paper. First, it separated the first-order rest energy μ from the quadratic invariant $\mu^2 = \det P$ instead of allowing one symbol to drift between operator orders. Second, exact unitarity exposed the finite-spacing corner ratio $\tan(a\mu)$, replacing the tempting but false finite identity $a\mu = \tan(a\mu)$. Third, proving an iterated product estimate separated a one-step Taylor expansion from genuine uniform many-step control. These are not the only bookkeeping gains. Fourth, the exact $3 + 1$ composition preserved the noncommuting $x/y/z$ /mass factor order instead of identifying reversed local updates. Fifth, the $1 + 1$ full-zone audit separated the zero-quasienergy cone from its high-quasienergy Floquet partner. Sixth, the $3 + 1$ high-symmetry audit disproved the tempting global reading of continuum isotropy: the tangent is isotropic, three nonzero corners exactly alias the origin, and at body center the massive step retains explicit $+1$ and -1 modes for every mass angle. Seventh, the exact Laurent residual ruled out the most economical proposed repair: a nonzero stationary amplitude cannot coexist with origin normalization, exact unitarity, and a full involutory tangent inside one degree-one nearest-neighbor factor. Eighth, the inhomogeneous phase calculation separated a global basis rotation from a genuine endpoint connection and proved that a global real lift cannot carry nonzero winding. Ninth, exact temporal blocking showed that the mass-only group law closes while the noncommuting spatial split generates a larger ordered-word family. Tenth, the radial derivative of the positive Pluecker action proved that its homogeneity collapses rather than selects a nonzero scale. These are the formalization contributions; the number of declarations is not.

9 What is forced, derived, realized, and open

The central scientific distinction is not between bold and cautious language. It is between different logical verbs.

Verb	Claim in this paper
Forced	A quadratic invariant vanishing on the full null boundary is proportional to the determinant; a nonnegative first-order gap is proportional to its square root.
Derived	The spinor pair gives z ; z gives B_z ; B_z gives the Dirac square, rest gap, unitary turn gate, and orientation-sensitive corner phase; a site field $z(x)$ gives its own patched link data with an exact 2π -integer winding.
Realized	The split-step null walk exhibits the construction exactly and converges uniformly on bounded $1+1$ momentum boxes; a finite local $3+1$ successive-axis walk has an ordered complex-Pluecker symbol with the same explicit rate, exact three-torus Plancherel control, and a classified high-symmetry failure set.
Open	A walk-specific changing-lattice Fourier/PDE limit, a replacement $3+1$ regulator that removes the corner aliases and body-center modes, beyond the excluded degree-one stationary-amplitude ansatz, reduction of the derived link data to a corrected non-winding-only defect index and stable localized mode, a graph-metric cone for the exact placed pair layer and its composition with the spatial free walk, an operator-exponential/action law, and a continuum interaction theory.

What can an assigned scalar mass mimic? Everything that depends on $|z|$ alone: after setting $m = |z|$, the constant-coefficient spectrum, the dispersion, the velocity bound, and the uniform many-step limit are identical, and the paper says so without embarrassment. What it cannot mimic is now a theorem list rather than a narrative. It has no input from which the rest operator is derived rather than declared (the classification theorem makes “derived” precise); it carries one real coin where the exact history sum carries z and $\bar{z}$ on opposite orientations; it produces no endpoint connection, no integer link winding, and no patched transition data from a varying phase unless that field is separately supplied; and a scalar mass, being one real number, cannot reproduce the exact $4/5$ -versus-1 interference split that separates two equal-modulus complex Pluecker fields — matching it requires importing exactly the complex datum this paper derives, together with the pair interaction that reads it. At constant z the distinction is structural: the economy of deriving mass and interaction phase from one object. Once the phase varies in space, the distinction becomes patched link data with an integer winding; whether the free walk forces the interaction that would make the constant- z phase a free-theory observable is open.

The paper proves a finite free construction and one finite phase-sensitive quartic extension. It does not prove that every massive particle is composite in the ordinary particle-physics sense. The “constituents” here are null momentum factors of a finite spinor description, not additional asymptotic particles. Nor does the theorem compute an absolute mass scale: rescaling the input spinors rescales z and hence μ . The quartic-action no-go sharpens this from an omission to a proved boundary for the natural positive homogeneous action.

The result does support a precise statement: a timelike rest gap can be encoded entirely by massless rank-one momentum primitives, and the gap is exactly the area opened between their spinor directions. In the derived walk, microscopic translation occurs along null channels while subluminal drift and a rest gap arise from coherent direction changes. This is an exact property of the finite model, not yet a universal ontology of nature.

10 Interpretation and open problems

The larger null-information program studies a four-term finite carrier square, positive cohomology, closure interactions, and discrete frame data. Those structures motivated the present theorem but

are not required to prove it. They belong in a companion paper unless they acquire a theorem that depends on B_z .

The next problems are sharply ordered.

1. **Complete the lattice-to-PDE identification.** The finite product DFT, its exact N^{-3} normalization, and the walk-specific compact-support momentum-space L^2 limit are closed. Control the changing lattice/continuum Fourier spaces and identify the limiting multiplier with the position-space Dirac PDE propagator. The abstract bulk/tail split and vanishing of the ultraviolet tail under measurable monotone band exhaustion are now closed for arbitrary L^2 data; what remains is the concrete scaled sampling/interpolation isometry and its composition with the walk symbol. Equations (31), (32), (33), and (34) close the finite, abstract countable, and compact-support multiplier steps, not that final physical identification.
2. **Remove the lattice partners without losing the derivation.** The no-go (48) rules out the naive degree-one stationary-amplitude factor with a full Dirac tangent. The Wilson construction (54) now removes the unwanted corners and gives a uniform massive gap for a nearest-neighbor Hamiltonian. The open task is to retain that spectral success in a strictly finite-range, exactly unitary discrete-time update. Test the enlarged internal cells, longer Laurent range, and tilted-projector stay-put constructions of the kind realized by Gupta and Short [13] while preserving the Pluecker-derived coin and exact path dictionary; by the corollary above, any such stay-put route must quantify what replaces the involutory unit-speed tangent. In $3 + 1$, remove the three proven even-parity origin aliases and both mass-independent body-center modes while retaining locality, unitarity, continuum isotropy, and the complex-mass rate, and eliminate the residual non-Dirac states that persist in the Gupta-Short family. Equations (40) and (41) are the regression tests.
3. **Complete the tetrahedral comparison.** Compute the tetrahedral walk's full Bloch symbol in a common blocked unit cell and audit its corners, bands, and locality radius. Constant onsite equivalence is already constrained by the alias-preservation no-go; a genuine comparison must specify any blocking or coarse-graining map.
4. **Generalize the Pluecker rest operator.** For larger null families, seek a canonical odd Hermitian operator from $w = (\psi_i \wedge \psi_j)_{i < j}$ whose square is the complete area budget and whose phase/moduli classification is decomposition-independent.
5. **Reduce the derived winding to an index and a localized mode.** The position-dependent unitary update, its inverse and causal cone, and exact site-dependent chiral connection are closed, and the derivation gap is now closed as well: the link increments of any nowhere-zero Pluecker field are the principal arguments of its neighbor ratios, their cycle sum is an exact 2π -integer winding, a globally lifted real phase still forces zero, and an explicit winding-one field is generated by primitive spinors. The next theorem must reduce this derived winding, or a sign/domain wall through the collinear zero set, to a finite index operator and exhibit a localized protected mode with a displayed stability class.
6. **Complete the many-body and interacting lift.** The generic finite CAR algebra, a phase-sensitive four-mode quartic pair-transfer observable, and an exact norm-preserving pair kick are now closed; the quartic operator is Hermitian and the kick/operator restriction identity is exact. The determinant-minor lift now fixes vacuum, agrees with the one-particle matrix, preserves number and parity, obeys exact creation covariance, and satisfies $\Gamma(1) = 1$ and $\Gamma(UV) = \Gamma(U)\Gamma(V)$. Its ordered-minor adjoint law, lifted two-sided inverse, and finite Fock

inner-product preservation are also exact. Annihilation covariance is now exact, and the placed quartic pair layer has exact disjoint-block commutation and finite scheduled CAR-support propagation. Add contiguous finite-range layers and a radius-by-time theorem, compose with the spatial one-particle walk, then derive an operator-exponential law and the quartic dynamics from one field-valued action whose reductions also produce the null-spinor geometry and one-particle evolution.

7. **Add a scale-bearing selection mechanism and derive couplings.** The positive homogeneous Pluecker action now has a proved no-go: every positive configuration has only the collapsed stationary common scale. Therefore a nonzero solution must display a competing homogeneity, constraint, ensemble scale, or other dimensionful input and quantify which observed scale remains supplied. Interaction strengths and gauge representations remain open.

The finite construction encodes spacetime information algebraically through a fixed Pauli map and null direction grading. Emergence of a dynamical continuum geometry remains conjectural. We therefore do not claim that the paper starts without spacetime structure; it starts without primitive timelike motion or a primitive mass term inside the finite model.

11 Conclusion

The result readers should retain is compact. Null spinors open an exterior area. That area is the positive rest gap, its square is the momentum determinant, and its complex orientation is the off-diagonal part of an odd Hermitian mass operator. The collinear null boundary is exactly the gap-closing locus.

The operator does more than rename a scalar. It produces the relativistic Dirac square, exponentiates to the exact unitary turn, and fixes the finite corner ratio, including the directed $z/\bar{z}$ factors in the exact history sum. The associated split-step walk converges uniformly on bounded $1 + 1$ momentum boxes at an explicit $O(1/n)$ rate, and its transfer-matrix history representation has a two-weight recursion that scales to the checkerboard corner ratio $-i \tan(a\mu)$. The finite local $3 + 1$ extension accepts arbitrary position-dependent Pluecker profiles, preserves norm exactly, has an explicit inverse and causal cone, and turns local phase variation into endpoint links. Its ordered symbol carries the complex phase at the same uniform compact-box rate, the three-torus Plancherel identity fixes the exact N^{-3} wave-packet normalization, and compactly supported momentum-space data converge in L^2 at every fixed time. The site field supplies its own patched link data with an exact integer winding, generated by primitive spinors, and the retained phase — invisible to the one-particle rest spectrum at constant modulus — is read out by an exact two-particle interference probability inside the supplied pair-kick sector. These statements form one finite chain from null geometry to rest energy and propagation. The same audit proves where that chain presently stops: the cubic regulator has three exact zero-quasienergy corner aliases, its massive body-center step retains exact 0 and π quasienergies for every mass angle, and for every intermediate mass angle those body-center-type points are provably the only zero- or π -quasienergy crossings. The degree-one stationary-amplitude no-go excludes the simplest attempted repair, and its factorized form proves that every onsite mass coin leaves all three even-parity aliases intact. A Wilson term removes the non-origin corners and opens a uniform massive gap at the local-Hamiltonian level; converting that success into a strictly finite-range discrete-time regulator remains open. Exact temporal blocking further shows that the mass subgroup closes while the full split family generates a larger ordered monoid, and the homogeneous-action no-go proves that a nonzero absolute mass needs an additional scale-bearing input.

What remains is equally definite: a changing-lattice position-space PDE limit, a local strictly finite-range unitary regulator outside the degree-one stationary-amplitude no-go that matches the Wilson Hamiltonian’s spectral success without sacrificing the derived coin, a blocked comparison with the tetrahedral construction, a patched defect/index theorem, and a second-quantized interacting extension. Success would extend a machine-verified finite rest-gap construction into a broader discrete relativistic dynamics. Failure at any named step would locate the boundary of the idea rather than dissolve it into analogy.

References

- [1] H. Elvang and Y.-t. Huang, *Scattering Amplitudes*, arXiv:1308.1697.
- [2] N. Arkani-Hamed, T.-C. Huang, and Y.-t. Huang, *Scattering Amplitudes For All Masses and Spins*, JHEP 11 (2021) 070, arXiv:1709.04891.
- [3] S. Dittmaier, *Weyl–van der Waerden formalism for helicity amplitudes of massive particles*, Phys. Rev. D 59 (1998) 016007, arXiv:hep-ph/9805445.
- [4] R. P. Feynman and A. R. Hibbs, *Quantum Mechanics and Path Integrals*, McGraw–Hill, 1965.
- [5] T. Jacobson and L. S. Schulman, *Quantum stochasticity: the passage from a relativistic to a non-relativistic path integral*, J. Phys. A 17 (1984) 375.
- [6] A. Bisio, G. M. D’Ariano, and A. Tosini, *Quantum field as a quantum cellular automaton: the Dirac free evolution in one dimension*, Ann. Phys. 354 (2015) 244, arXiv:1212.2839.
- [7] G. M. D’Ariano, N. Mosco, P. Perinotti, and A. Tosini, *Path-integral solution of the one-dimensional Dirac quantum cellular automaton*, Phys. Lett. A 378 (2014) 3165–3168, arXiv:1406.1021.
- [8] A. Mallick and C. M. Chandrashekar, *Dirac cellular automaton from split-step quantum walk*, Sci. Rep. 6 (2016) 25779, arXiv:1509.08851.
- [9] S. Succi, F. Fillion-Gourdeau, and S. Palpacelli, *Quantum lattice Boltzmann is a quantum walk*, EPJ Quantum Technol. 2 (2015) 12.
- [10] A. Bisio, G. M. D’Ariano, P. Perinotti, and A. Tosini, *Free quantum field theory from quantum cellular automata: Derivation of Weyl, Dirac and Maxwell quantum cellular automata*, Found. Phys. 46 (2016) 1032–1081, arXiv:1601.04832.
- [11] U. Nsongani, N. Eon, I. Marquez-Martin, A. Perez, G. Di Molfetta, and P. Arrighi, *Dirac quantum walk on tetrahedra*, Phys. Rev. A 110 (2024) 042418, arXiv:2404.09840.
- [12] J. K. Asboth and H. Obuse, *Bulk-boundary correspondence for chiral symmetric quantum walks*, Phys. Rev. B 88 (2013) 121406(R), arXiv:1303.1199.
- [13] C. Gupta and A. J. Short, *Fermion doubling in Dirac quantum walks*, Phys. Rev. A, accepted (2026), arXiv:2601.15885, doi:10.1103/kkz3-fthm.
- [14] L. Mlodinow and T. A. Brun, *Discrete spacetime, quantum walks, and relativistic wave equations*, Phys. Rev. A 97 (2018) 042131, arXiv:1802.03910.

- [15] A. M. Childs, Y. Su, M. C. Tran, N. Wiebe, and S. Zhu, *A theory of Trotter error*, Phys. Rev. X 11 (2021) 011020, arXiv:1912.08854.
- [16] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, CADE-28, Lecture Notes in Computer Science 12699 (2021) 625–635.
- [17] The Mathlib Community, *The Lean mathematical library*, Proceedings of CPP 2020, 367–381.
- [18] J. Tooby-Smith, *HepLean: Digitalising high energy physics*, arXiv:2405.08863.
- [19] M. R. Douglas, S. Hoback, A. Mei, and R. Nissim, *Formalization of QFT*, arXiv:2603.15770.